

System Requirements Specification (SRS)

Project Title: No-Registration Web-Based 2D Food Packaging Editor

Target Platform: Web (Desktop & Mobile-responsive browser)

1. Key User Workflow & System Behavior (No-Registration Model)

Since users do not need to register or log in, the system handles saving and session persistence using local storage and file downloads:

1. **Guest Session Initialization:** When a user visits the app, a temporary session ID is created automatically.
2. **Local Browser Saving:** Project states are saved automatically to the browser's `localStorage` / `IndexedDB`. If the user accidentally refreshes, their work remains intact.
3. **File-Based Project Saving & Loading:**
 - **Save:** Users click "Save Project" to download a lightweight `.JSON` project file directly to their computer.
 - **Open/Resume:** Users can upload an existing `.JSON` project file to pick up right where they left off without needing an account.

2. Functional Requirements

2.1 Food Packaging Template Library

The system will feature pre-built parametric templates tailored specifically for the food and beverage industry:

- **Takeout & Bakery:** Lock-bottom cake/donut boxes, burger clamshells, French fry scoops, takeaway handle boxes.
- **Pouches & Bags:** Stand-up food pouches (ziplock style), coffee bean bags, paper sandwich bags.
- **Containers & Sleeves:** Meal-prep sleeve bands, pizza boxes, popcorn tubs, cup sleeves.
- **Custom Dimension Inputs:** Users can adjust base internal dimensions: Length (\$L\$), Width (\$W\$), Depth (\$D\$), and Paper/Cardstock Thickness (\$t\$).

2.2 Interactive Side-by-Side Face Mapping & Canvas

- **Panel Highlighting:** The 2D dieline clearly labels distinct packaging faces (*Front, Back, Top Lid, Bottom, Left Side, Right Side, Lock Flap*).
- **Face Selection & Focus:** Clicking on a specific side (e.g., "Front Panel") isolates or snaps the canvas focus directly to that face for easy editing.

2.3 Side-Level Graphic & Text Editing

Users can customize each face using the following tools:

- **Image & Logo Uploads:**
 - Drag and drop PNG, JPG, WebP, or SVG graphics directly onto any panel.

- System auto-crops or clips artwork so it does not spill outside the panel bounds (unless set to Bleed).
- Layer positioning: Bring Forward, Send Backward, Scale, Rotate, and Align (Center, Left, Right).
- **Text Engine:**
 - Add custom text fields (e.g., Brand Name, Ingredients List, Nutrition Facts, Expiration Date notices).
 - Text controls: Font selection, size, color picker, bold, italic, line height, alignment, and curved text option for round food tubs.
- **Preset Graphics & Icons:**
 - Built-in library of vector food compliance icons (*Recyclable, Keep Frozen, Microwave Safe, Halal, FDA Approved, Organic, Storage Instructions*).
- **Barcode & QR Code Generator:**
 - Auto-generate vector QR codes (linking to menus, social media, or promo sites) and standard barcodes (EAN-13, UPC) directly onto the design.

2.4 Export & Save Engine

- **Save / Load Project Data:**
 - Export working state as a **.JSON** file (e.g., *my-burger-box-design.json*).
 - Upload a **.JSON** file to restore artwork, layers, dimensions, and text instantly.
- **Production Print Export (PDF & SVG):**
 - Export vector PDF / SVG formatted for professional print shops.
 - Separate layer visibility: **Cut Lines** (Red), **Fold Lines** (Dashed Green), **Artwork** (Full Color CMYK print layer).
- **High-Res Quick Image Export:**
 - Export flat 2D renders as PNG/JPG (\$300\text{ DPI}\$) for quick previewing or printing on home/office desktop printers.

3. Architecture & Tech Stack Recommendation

Layer	Technology	Why It Fits
Frontend Framework	React.js / Next.js	Fast UI rendering, component modularity
2D Canvas Engine	Fabric.js or Konva.js	Handles images, text objects, selection handles, and SVG clipping natively

Dieline Geometry Engine	Paper.js / Custom TypeScript	Calculates vector paths for folding boxes based on \$L \times W \times D\$
Client Storage	LocalStorage + LocalForage (IndexedDB)	Retains draft work without requiring backend database user profiles
Export Engine	jspdf & makerjs	Generates layer-separated PDF and DXF vector files right inside the browser

4. Non-Functional Requirements

- **Zero Backend Friction:** No login screens, email verification, or passwords. Instant access to the editor.
- **Client-Side Processing:** All image uploads and layout manipulations execute directly on the user's browser, ensuring high privacy and near-instant performance without server bottlenecks.
- **Responsive Canvas:** Canvas supports pinch-to-zoom and touch panning for tablet and desktop users.

5. Simplified MVP Development Plan

1. **Phase 1 (Template Engine & Canvas):** Create 3-5 core food templates (Burger Box, Pizza Box, Stand-up Pouch) using Fabric.js.
2. **Phase 2 (Editor Controls):** Build image/logo upload buttons, text controls, and the food icon library.
3. **Phase 3 (Save/Load State):** Implement JSON export/import functionality and LocalStorage auto-save.
4. **Phase 4 (Print Export):** Build high-resolution PNG and print-ready vector PDF export functions.